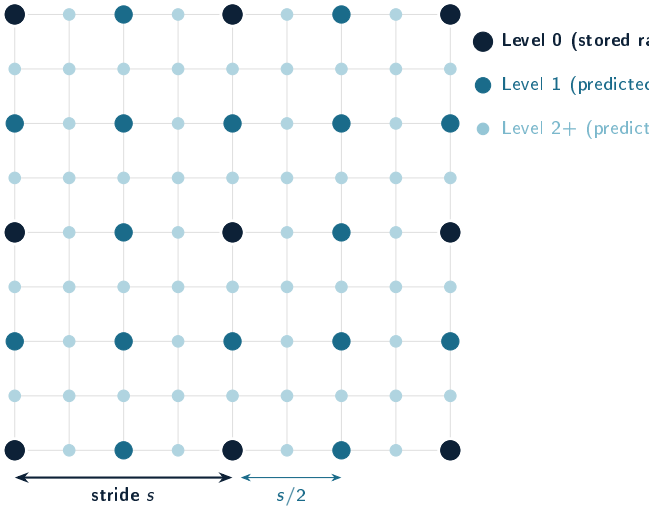
Dyadic Grid Refinement in QoZ / SZ3

QoZ and **SZ3** with `ALGO_INTERP` use a **dyadic (halving) refinement** schedule to predict data values hierarchically.

- **Level 0** — coarsest grid
  Points spaced at initial stride (e.g. 32).
  Stored as raw floats — no prediction.
- **Level 1** — halved stride
  Midpoints predicted by cubic spline interpolation from Level 0 anchors.
- **Level 2+** — repeat halving
  Each level predicts from the previous, recursively refining the grid.

Prediction residuals are quantized and entropy-coded (Huffman + zstd). Anchor points dominate the bit budget

at low error bounds.



GP Pre-Pass: Replacing Cubic Splines at L0/L1

Why a GP instead of a cubic spline?

At L0/L1, anchor points can be *far apart* (stride = 32+). Cubic splines use only the **nearest 4 neighbors —** they extrapolate poorly across large gaps and cannot exploit smooth global structure. A GP with a stationary kernel (e.g. Matérn-$\frac{5}{2}$) conditions on *all* stored anchors simultaneously, producing a posterior mean that remains accurate even when the prediction point is far from every anchor. It also degrades gracefully: uncertainty (σ) grows with distance, while the mean still exploits the global covariance structure.

GKS anchor selection [?]:

Greedy algorithm: pick the subset of L0 grid points that minimizes worst-case GP posterior variance over the domain. Those points are stored at *full precision*; all others are predicted.

- **GKS anchors —** stored raw GP training inputs; $\sigma = 0$ here.

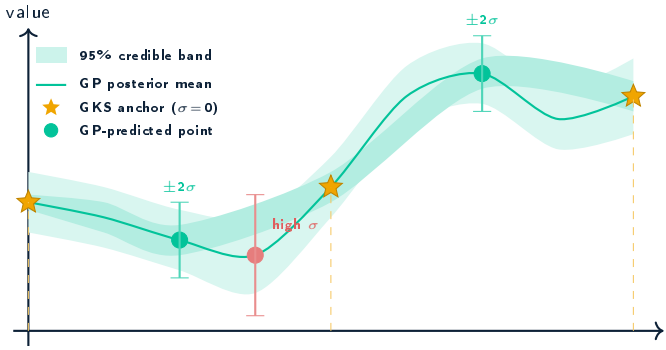


Savings: fewer stored anchors ⇒ smaller header; GP captures smooth spatial structure missed by local splines.

Quantified Uncertainty in GP-Predicted Anchors

Unlike cubic splines, a GP returns not only a **posterior mean** $\mu(x)$ but also a **posterior variance** $\sigma^2(x)$ at every predicted point.

- **Anchor points** (GKS) fix $\sigma = 0$
  GP is conditioned on these; uncertainty collapses.
- **Predicted anchors** carry $\sigma(x)$
  Uncertainty grows between stored anchors and
  shrinks near them — shown right.
- **Adaptive quantization**
  σ can guide EB allocation: tighter bounds where the GP is uncertain, looser where it is confident.

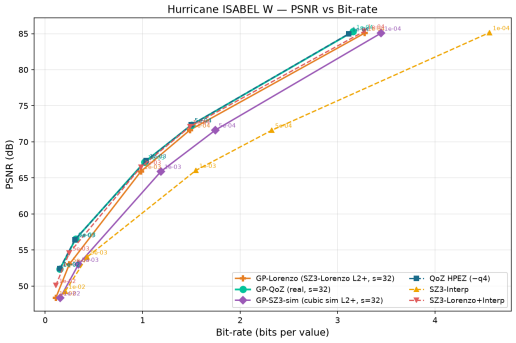


Potential use

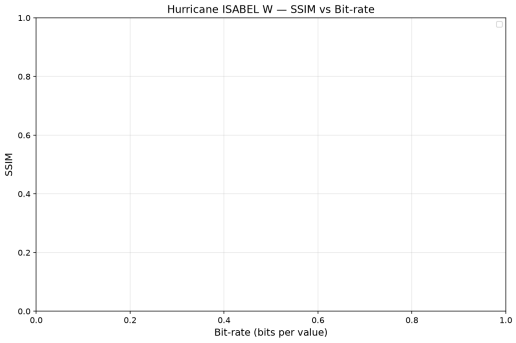
Rate–distortion optimization: allocate more bits at high-σ predicted anchors, fewer bits at low-σ

Hurricane ISABEL W — Rate-Distortion Results

PSNR vs. Bit-rate



SSIM vs. Bit-rate



Dataset: Hurricane ISABEL `Wf48` ($100 \times 500 \times 500$ f32). GP-QoZ: stride $= 32$, `gp_levels` $= 2$, Huffman + zstd. **Labels next to each point** show the relative error bound ε tested at that setting ($\varepsilon \in \{10^{-2}, 5\times10^{-3}, 10^{-3}, 5\times10^{-4}, 10^{-4}\}$; smaller ε = tighter bound = higher bit-rate).